HU Collection
TMUA Mock Set C Paper 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	E	D	C	F	B	C	A	D	B	C

Q	11	12	13	14	15	16	17	18	19	20
Ans	A	F	B	A	E	D	E	F	D	F

Each solution gives an intended route and names the trap the question was built around. Diagrams show the configuration each solution works with.

Question 1. Answer **E**.

The statement is "if it does not snow, the train is on time". An implication is false in exactly one situation: its hypothesis holds and its conclusion fails. So the negation is "it does not snow and the train is not on time". Option D is the contrapositive, which is equivalent to the original, not its negation. Option F is "not(not snow) or not on time" with the first negation dropped, and in any case the negation of an implication is never an "or".

Question 2. Answer **D**.

"If P then Q" is false only when P is true and Q is false. Then I has a false hypothesis (not P), so it is true. II has a false hypothesis (Q), so it is true. III has a true hypothesis and a true conclusion (not Q), so it is true. IV is the contrapositive of the original, and has the same truth value: false. The single fact to hold on to is that $P \Rightarrow Q$ is equivalent to (not P) or Q, so its negation is P and (not Q).

Question 3. Answer **C**.

Write each as a disjunction. The original is (not P) or Q or R. I is not(P and not Q) or R, which is (not P) or Q or R: equivalent. II is the contrapositive of the original: equivalent. III is ((not P) or Q) or ((not P) or R), which simplifies to (not P) or Q or R: equivalent. IV is ((not P) or Q) and ((not P) or R), which is (not P) or (Q and R): stronger, and false when P and Q are true and R is false.

Question 4. Answer **F**.

"Increasing" is: for all x, y, (if $x < y$ then $f(x) < f(y)$). Negating swaps the quantifier and negates the implication, and the negation of "if A then B" is "A and not B": there exist x, y with $x < y$ and $f(x) \geq f(y)$. Option B is the classic mistake of negating only the conclusion. It is true for every function whatsoever, since choosing $x = y$ makes the hypothesis $x < y$ false and the implication vacuously true.

Question 5. Answer **B**.

A proof by contradiction of "for all x, if $A(x)$ then $B(x)$" starts from its negation: there is an x with $A(x)$ and not $B(x)$. Here $B(x)$ is $1 < x < 2$, meaning $x > 1$ and $x < 2$, and its negation is $x \leq 1$ or $x \geq 2$. Option C negates the conclusion incorrectly: no x satisfies both $x \leq 1$ and $x \geq 2$, so that assumption leads to a contradiction for a trivial reason and proves nothing about the statement.

Question 6. Answer **C**.

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A counterexample needs the hypothesis true and the conclusion false. At $\frac{\pi}{6}$ and $\frac{2\pi}{3}$ the hypothesis holds and $\sin x + \cos x > 0$. At $\frac{5\pi}{6}$ the hypothesis holds ($\sin x = \frac{1}{2}$) and $\sin x + \cos x = \frac{1-\sqrt{3}}{2} < 0$: a counterexample. At $\frac{7\pi}{6}$ both are negative, so the hypothesis fails and this is not a counterexample, however negative the sum. At $\frac{5\pi}{3}$, $\cos x = \frac{1}{2} > 0$ and the sum is $\frac{1-\sqrt{3}}{2} < 0$: a counterexample. At $\frac{7\pi}{4}$, $\cos x > 0$ and the sum is exactly 0, which is not > 0: a counterexample. Three in total.

Question 7. Answer **A**.

The student assumes n odd and deduces n^3 odd, which is the converse: if n is odd then n^3 is odd. The statement asked for is nevertheless true, and the right proof is by contrapositive: if n is even, $n = 2k$, then $n^3 = 8k^3$ is even. The contrapositive of "n^3 odd $\Rightarrow$ n odd" is "n even $\Rightarrow$ n^3 even", not "n odd $\Rightarrow$ n^3 odd".

Question 8. Answer **D**.

"A if B" means $B \Rightarrow A$; "A only if B" means $A \Rightarrow B$. I is $x^2 > 4 \Rightarrow x > 2$: false at $x = -3$. II is $x > 2 \Rightarrow x^2 > 4$: true. III is true. IV is $x^3 > 8 \Rightarrow x > 2$: true, since cubing is increasing. V is $x^3 > 8 \Rightarrow x^2 > 4$: true, via $x > 2$. VI is $x^2 > 4 \Rightarrow x^3 > 8$: false at $x = -3$. Four are true.

Question 9. Answer **B**.

$\cos 2x < \frac{1}{2}$ is $1 - 2\sin^2 x < \frac{1}{2}$, that is $\sin^2 x > \frac{1}{4}$, or $|\sin x| > \frac{1}{2}$. So $\sin x > \frac{1}{2}$ implies it, and is sufficient. It is not necessary: $x = -\frac{\pi}{2}$ has $\cos 2x = -1 < \frac{1}{2}$ but $\sin x = -1$.

Question 10. Answer **C**.

I: $\sin x + \cos x = \sqrt{2}\sin(x + \frac{\pi}{4}) \leq \sqrt{2}$. II: $\sin x \cos x = \frac{1}{2}\sin 2x \leq \frac{1}{2}$. III: $\sin^4 x + \cos^4 x = (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = 1 - \frac{1}{2}\sin^2 2x \geq \frac{1}{2}$. IV fails at $x = \frac{5\pi}{4}$, where the sum is $-\sqrt{2}$. The lower bound matching I is $\geq -\sqrt{2}$, not -1.

Question 11. Answer **A**.

The student handles $\cos x = 0$ separately, where the inequality reads $0 > 0$ and fails, and elsewhere divides by $\cos^2 x > 0$, which is legitimate and preserves the direction. So the original inequality is equivalent to $\tan x > 1$ with $\cos x \neq 0$, whose solution set in $[0, 2\pi)$ is exactly the one given. Compare this with dividing $\sin x > \cos x$ by $\cos x$, where the sign of $\cos x$ matters: the difference is that $\cos^2 x$ is never negative.

Question 12. Answer **F**.

I: $(\sin x + \cos x)^2 = 1 + \sin 2x > 1$ on this interval, and the sum is positive. II: $\sin 2x = 2\sin x \cos x < 2\sin x$ since $0 < \cos x < 1$ and $\sin x > 0$. III: $\cos 2x - \cos x = 2\cos^2 x - \cos x - 1 = (2\cos x + 1)(\cos x - 1) < 0$ here, so III is false for every such x. IV: $\tan x = \frac{\sin x}{\cos x} > \sin x$ since $0 < \cos x < 1$. So I, II and IV.

Question 13. Answer **B**.

All four together mean $8 \mid n$ and $3 \mid n$, that is $24 \mid n$. Remove (1): $6, 12, 8 \mid n$ still gives $24 \mid n$. Remove (2): $4, 12, 8$ gives 24. Remove (3): $\text{lcm}(4, 6, 8) = 24$. Remove (4): $\text{lcm}(4, 6, 12) = 12$, so $n = 12$ now slips in. Only (4) is not removable. Note that removability is about one statement at a time: (1), (2) and (3) are each removable, but not all three together.

Question 14. Answer **A**.

C: $8 \mid n^2$ forces n even; writing $n = 2m$, $8 \mid 4m^2$ means m is even, so $4 \mid n$. Conversely $4 \mid n$ gives $16 \mid n^2$. So $A \Leftrightarrow C$. D: $12 \mid n^2$ means $4 \mid n^2$ and $3 \mid n^2$, that is $2 \mid n$ and $3 \mid n$, so $6 \mid n$; conversely $6 \mid n$ gives $36 \mid n^2$. So $B \Leftrightarrow D$.

Question 15. Answer **E**.

Count true statements region by region. For $x > 1$ all five hold. For $-1 < x \leq 1$, $x > -1$ holds, and $x^2 > x$ holds only when $x < 0$; so two are true on $(-1, 0)$ and one on $[0, 1]$. At $x = -1$ only $x^2 > x$

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holds. For $x < -1$, exactly $x^2 > 1$ and $x^2 > x$ hold. So exactly two are true for $x < -1$ and for $-1 < x < 0$, but not at $x = -1$ itself.

Question 16. Answer **D**.

I: $4ac > b^2 \geq 0$ and $a > 0$ force $c > 0$. II fails: $a = b = 0$, $c = 1$ gives $f \equiv 1 > 0$, yet (1) fails. The question does not say f is quadratic, and that is the whole trap. III: $4ac > b^2 \geq 0$ with $c > 0$ forces $a > 0$. So I and III.

Question 17. Answer **E**.

All four together are equivalent to $a > 0$ and $b > 0$. Test the six pairs: (1)(2): $ab > 0$ means same sign, and a positive sum makes both positive; equivalent. (3)(4): trivially equivalent. (2)(3): $a > 0$ and $ab > 0$ give $b > 0$; equivalent. (2)(4): likewise. (1)(3): $a = 5$, $b = -1$ satisfies both but not (4). (1)(4): symmetric. So four pairs. Every single condition is also individually removable here, which shows that "removable one at a time" and "which small subsets suffice" are different questions.

Question 18. Answer **F**.

I: $f(x) \leq f(1)$ throughout, so the integral is at most $f(1)$. III: substituting $x \mapsto x + \frac{1}{2}$ gives $\int_{1/2}^{1} f = \int_0^{1/2} f(x + \frac{1}{2})\,dx \geq \int_0^{1/2} f(x)\,dx$. II is the one needing an idea. Since $\int_0^1 (x - \frac{1}{2})\,dx = 0$,

$$\int_0^1 \left(x - \tfrac{1}{2}\right) f(x)\,dx = \int_0^1 \left(x - \tfrac{1}{2}\right)\left(f(x) - f\left(\tfrac{1}{2}\right)\right)dx \geq 0,$$

because both brackets are negative for $x < \frac{1}{2}$ and non-negative for $x > \frac{1}{2}$. That rearranges to II. All three.

Question 19. Answer **D**.

I: f is odd. II: the substitution $x \mapsto a - x$ works for any f. IV: with $u = x - a$ the integral is $\int_{-a}^{a} f(u)\,du = 0$. III fails: $xf(x)$ is even, not odd, and $f(x) = x$ gives $\int_{-a}^{a} x^2\,dx > 0$. So I, II and IV.

Question 20. Answer **F**.

From $S(10)$, the second rule gives $S(8), S(6), S(4), S(2)$. Now combine the rules: if $S(n)$ holds then $S(n + 3)$ holds, and then the second rule gives $S(n + 1)$. So each true statement makes the next one true, and once $S(1)$ is reached everything follows. It is: $S(2) \Rightarrow S(5)$, then $S(5) \Rightarrow S(3) \Rightarrow S(1)$. So $S(n)$ is true for every positive integer n. Seeing that $+3$ followed by -2 is a step of $+1$ is the idea; tracking individual implications by hand, you tend to stop at one of the partial sets in the other options.

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